

SANDESH CHAPAGAIN

Backend Engineer · Co-founder · Rome, Italy ✉️sendmailtosex@gmail.com · 📞 (+39) 344 594 6149 · 🌐 iamdex.xyz

PROFILE

Founded Ditch : a production live-streaming architecture with a custom RTMP relay engine, WebRTC participant pipelines, canvas architecture compositor, and real-time multi-platform broadcast. Built the complete technical stack from scratch while enrolled in Engineering Sciences at Tor Vergata. Accepted into Startup Lazio 2026. Looking for early-stage infrastructure and product engineering roles.

PROJECTS

Ditch — Cross-Platform Live-Streaming Architecture

Jan 2025 – Present

Co-founder & Sole Engineer · Startup Lazio 2026 · Tor Vergata

- Custom RTMP relay engine in Node.js: one browser ingest fans out simultaneously to YouTube, Twitch, Kick, and TikTok Live via ffmpeg tee muxer. Stream encoded once, pushed everywhere.
- Canvas compositor at stable 30fps with tab backgrounded — Web Worker frame timer (Workers bypass browser throttling) + Breakout Box API + VideoFrame compositing. Layout adapts dynamically: solo, side-by-side, split-row, 2x2 grid.
- Real-time server: Socket.io handling unified chat aggregation from all connected platforms, participant signaling, and live session state broadcast to all viewers.
- Backend: Fastify REST API, Supabase Postgres, participant state machine, server-side RTMP credential management.
- Monorepo: Turborepo + pnpm, 4 apps (web, api, realtime, relay), 2 shared packages (types, db). Frontend on Vercel, API / realtime / relay on Railway.

Next.js TypeScript Fastify Socket.io WebRTC Agora SDK ffmpeg Canvas API Web Workers Supabase Railway Turborepo

RTMP Stream Tools — Internal CLI Toolkit

2025

Built during Ditch relay development

- rtmp-probe: wraps ffprobe to inspect codec, bitrate, resolution, and FPS of any live RTMP stream.
- loopback-broadcaster: pushes a synthetic test stream (color bars + timecode overlay, or a video file) to any RTMP endpoint via ffmpeg — full relay pipeline testing without a live browser session.
- stream-health: pings RTMP destinations on an interval, logging reconnect events. Used daily during backend development.

Node.js TypeScript ffmpeg Node Media Server Commander.js

WebRTC from Scratch

2024 – 25

Peer-to-peer video calling without SDK dependencies

- Implemented ICE negotiation, STUN/TURN, and SDP offer/answer exchange in vanilla JS with a Node.js signaling server — no Agora, no Twilio.
- Key finding: most WebRTC connection failures trace to STUN misconfiguration. Directly informed Ditch's WebRTC architecture and reduced participant join failures in production.

JavaScript WebRTC Node.js STUN/TURN WebSockets

BarcaBuzz — Football Content Community

Aug 2022 – Jun 2023

Co-creator & Growth Lead

- Grew to 100K+ followers organically in under 10 months, zero paid spend.
- Built a structured content pipeline: per-post tracking, A/B testing across format variables (hook, caption length, thumbnail, posting time), engagement feedback loops fed directly back into content decisions.
- Key insight: save rate predicts long-tail reach better than immediate engagement. Optimizing for the right metric changes what you build.

Content Systems Analytics Audience Growth

WORK EXPERIENCE

Network Support Intern

Intrasoft Networking Solutions · Lalitpur, Nepal

Jun 2024 – Nov 2024

- Diagnosed and resolved network connectivity issues across client systems; monitored performance using Wireshark and Nagios; configured routers, switches, VoIP phones across multiple sites.

Marketing & Growth Manager

Pathik Gyan Niketan · Kathmandu, Nepal

Jul 2023 – Jun 2024

- Managed digital presence across Facebook, Instagram, and TikTok; ran Meta Ads and Google Ads campaigns with ROI tracking via Google Analytics.

EDUCATION

Bachelor in Engineering Sciences

Università degli Studi di Roma Tor Vergata · Rome, Italy

Dec 2024 – Present

- Mathematics · Physics · Electronics · Computing · Control Systems · Mechanics

Fundamentals of Digital Marketing

Google Skillshop · Remote

Jan – Feb 2023

School Leaving Certificate

Valmiki Shiksha Sadan · Bharatpur, Nepal

Aug 2020 – Jul 2022

TECHNICAL SKILLS

LANGUAGES	JavaScript (ES6+) · TypeScript · HTML5 · CSS3 · Python (foundational) · Golang (foundational)
REAL-TIME	WebRTC · RTMP · ffmpeg · Agora Web SDK · Node Media Server · WebSockets
FRAMEWORKS	Next.js · React · Node.js · Fastify · Tailwind CSS · Express
INFRA	Docker · Supabase · Railway · Vercel · Git · WSL2/Ubuntu · Turborepo
NETWORKING	TCP/IP · Wireshark · Nagios · Windows Server · Active Directory
ANALYTICS	Google Analytics · Meta Ads Manager · Google Ads

LANGUAGES

Nepali (Native) · English (C2) · Hindi (C2) · Italian (B1)